

Anchored multi-omics integration and knowledge-anchored residual discovery: a never-below-the-best-view integrator that finds what the textbook misses

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Running title: Knowledge-anchored residual discovery

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Abstract

Background. Multi-omics integration is widely assumed to improve prediction, yet on real cancer data a strong single modality is hard to beat and naive fusion often underperforms it. **Methods.** We define an *anchored* integrator that anchors on the empirically strongest modality — or on a fixed, zero-parameter textbook prior — and adds the remaining data only as a non-negative gated residual, so the model is provably never below its anchor and improves on it only where another view carries orthogonal signal. Mining that residual yields a discovery procedure that names anchor-orthogonal features, with a matched-random-panel noise control and held-out / permutation verification. We evaluate on TCGA-BRCA (n up to 1,152) and validate externally on METABRIC (n = 1,175–1,980). **Results.** Across nine subtype labels the gate adds nothing where one modality dominates and engages only on a constructed positive control (+0.047 AUROC). Anchoring on a fixed 20-gene proliferation signature (zero trained parameters) reaches AUROC 0.919 on Luminal A vs B — nearly the trained 1,500-gene model (0.942) — and gating genome-wide data reaches 0.947, the only clean super-additive gain observed. The residual of that prior recovers a basal/keratinization axis (Reactome $p \approx 8 \times 10^{-11}$), verified by held-out replication (10/10 splits), stability (10/10), and a permuted-label null; it **reproduces in METABRIC** (panel $\Delta +0.036$; independent re-discovery overlaps 20/30, hypergeometric $p \approx 7 \times 10^{-27}$). A second anchor (the ERBB2 amplicon for HER2) yields a distinct neuroendocrine/immune axis, and an estrogen-receptor signature yields nothing (a specificity control). The HER2 axis does **not** replicate in METABRIC because the amplicon anchor is near-complete there (0.997). The method is also not breast-cancer- or expression-specific: on an NSCLC anti-PD-1 immunotherapy cohort (n = 227, mutation/clinical features), anchored on the textbook biomarker tumour mutational burden, the residual independently recovers the field’s other established biomarkers — PD-L1 (orthogonal to TMB), EGFR and

STK11 mutation (resistance) (panel $\Delta +0.061$ vs $+0.006$, $p = 0.038$). Finally, the discovered axis replicates across *cancers*: in TCGA lung (LUAD vs LUSC, $n = 1,129$), the same proliferation-anchored residual recovers the squamous/keratinization program, overlapping the breast basal panel 10/30 ($p \approx 3 \times 10^{-16}$) — the same genes in a different cancer. **Conclusion.** Anchor on established biology and let a gate decide, honestly, whether genome-wide data beats it; where it does, the residual names the new axis — one that reproduces across cohorts (METABRIC) and across cancers (lung), and generalises to a different disease and feature type (NSCLC immunotherapy). External reproducibility tracks biological coherence.

1. Introduction

The prevailing premise that adding omics layers improves prediction is contradicted by careful benchmarks: a large-scale TCGA benchmark across 14 cancers found that adding data types most often *hurt* survival prediction, with mRNA ($\pm$ miRNA) sufficient for most cancers [1]; multimodal survival models often fail to beat a small two-modality subset, so the useful combination is task-specific and narrow [2]; and naive concatenation underperforms — a principal-component baseline is hard to beat, so adaptive weighting is needed to realise any gain from fusion [3,4]. The honest objective is therefore not “always fuse” but to *never do worse than the best single modality, and improve on it only where another modality carries signal the leader cannot*. We formalise such an integrator, generalise its anchor from a data modality to established knowledge, and repurpose the construction as an interpretable discovery engine — then ask, in an independent cohort, whether its discoveries reproduce.

2. Results

An anchored gate is never below the best single view. On Luminal A vs B, RNA alone reaches AUROC 0.947 and methylation 0.745; a naive RNA+methylation stack drops to 0.941 (below RNA), whereas the gated combiner holds at 0.947 (weight $\beta = 0$ in 9/10 repeats). A constructed positive control — a held-out methylation set the RNA cannot see — engages the gate ($\beta = 4-8$) for a significant $+0.047$. Run blind on nine expert-defined subtype labels, anchor selection routes to methylation for all five methylation-defined clusters and to RNA for all four PAM50 calls, never below the leader: the method follows the biology rather than a built-in preference.

A zero-parameter textbook prior is a strong anchor, and yields the only clean fusion gain (Figure 1A). Luminal A vs B is, by textbook, a proliferation distinction. A 20-gene proliferation index with zero trained parameters reaches AUROC 0.919 — close to the fully trained 1,500-gene model (0.942) — and gating genome-wide data onto this fixed prior reaches 0.947 ($\Delta +0.029$), exceeding the pure data model. With the Horvath epigenetic clock [9] as the anchor for normal-tissue age, the clock alone (0.947) already beats RNA (0.911) and data adds ≈ 0 : the textbook suffices and the gate says so.

The residual names a verified new axis. Mining the proliferation-anchored residual surfaces the basal / squamous-lineage axis (KRT5/14/17/6B, TP63, DSG3/DSC3, SOX10, COL17A1, KLK5/7/8; partial $r \approx 0.46$). It is verified three ways — a train-selected panel beats random panels on a held-out test in 10/10 splits (no selection leakage), the basal core recurs in 10/10 splits, and under permuted labels the advantage collapses — and is enriched for cornified-envelope formation / keratinization / epidermis development (Reactome $p \approx 8 \times 10^{-11}$).

Generalisation and specificity (Figure 1A). Anchoring HER2 status on the ERBB2 amplicon (incomplete in TCGA, AUROC 0.752) discovers a distinct, verified neuroendocrine/secretory + immune axis ($\Delta +0.054$; panel vs random $p = 0.024$; held-out 8/8). Anchoring ER status on a textbook ER/luminal signature (complete, 0.938) discovers nothing ($\Delta -0.001$): a real-data specificity control showing the method does not manufacture axes.

External validation (Figure 1B). In the independent METABRIC cohort the basal axis reproduces: the fixed TCGA panel adds $\Delta +0.036$ over the proliferation prior (combined 0.960; beats random panels, $p = 0.048$), and an unbiased re-discovery independently recovers the same basal genes, overlapping the TCGA panel 20/30 (hypergeometric $p \approx 7 \times 10^{-27}$). The HER2 axis does **not** reproduce — there the amplicon anchor is near-complete (0.997 vs 0.752 in TCGA), leaving no residual (panel $\Delta \approx 0$; overlap 2/30). External reproducibility tracks biological coherence: the pathway-enriched basal axis reproduces, while the diffuse HER2 axis was a cohort-specific residual.

Cross-domain generalisation (a different cancer, question, and feature type). The method is not specific to breast cancer or to expression. On NSCLC patients receiving anti-PD-1 checkpoint blockade (Hellmann/MSK 2018, $n = 227$; mutation and clinical features; endpoint = durable clinical benefit), anchoring on the textbook immuno-oncology biomarker tumour mutational burden (TMB; anchor AUROC 0.60) and mining the residual independently recovers the field’s *other* established biomarkers: PD-L1 score (positive partial correlation, and genuinely orthogonal to TMB, $\text{corr } 0.00$), EGFR mutation and STK11/LKB1 mutation (both negative — known checkpoint-blockade resistance). The discovered panel adds $\Delta +0.061$ versus $+0.006$ for matched random panels ($p = 0.038$). Anchored on the textbook biomarker, the procedure rediscovers the known complementary biomarkers of a different disease.

Cross-cancer replication and tissue-independence of the basal axis. The strongest test that a discovered axis is real biology, not a cohort artefact, is to seek it in a *different cancer*. In TCGA lung (LUAD adeno vs LUSC squamous, $n = 1,129$), the same proliferation-anchored residual recovers the squamous/keratinization program and overlaps the breast basal panel 10/30 (hypergeometric $p \approx 3 \times 10^{-16}$) — the same genes, a different cancer. (On this near-trivial histology split the panel-vs-random margin saturates, so gene-level overlap is the informative metric.) Adding head & neck (HNSC), the breast panel separates squamous (HNSC + LUSC) from adeno (LUAD) at AUROC 0.96 with HNSC and LUSC — two different tissues — both scoring high and LUAD low, and within HNSC the score tracks differentiation grade ($G1 > G3$, $p \approx 2 \times 10^{-4}$): the axis is a tissue-independent squamous-differentiation marker.

Clinical significance: identity, not outcome (an honest negative). A reproducible axis need not be prognostic. In TCGA-BRCA ($n = 866$, 132 events) the basal score is not associated with overall survival (Cox HR 1.01, $p = 0.89$; adjusted for proliferation $p = 0.36$; KM log-rank $p = 0.29$), whereas the proliferation score is ($p = 0.001$). The basal axis does mark ER-negative/basal-like disease (AUROC 0.70). The discovered axis therefore captures lineage *identity* rather than outcome — reported plainly, because the method’s value is finding real, robust biology, not inflated clinical claims.

Hypothesis-as-anchor — confirming, explaining-away, or refuting a hypothesis (Figure 2). The frame also inverts cleanly: a *hypothesis* is expressed as a candidate anchor and tested against the textbook anchor on real data, gating it onto the textbook residual for a three-way verdict — SUPPORTED (adds beyond the textbook), EXPLAINED_BY_TEXTBOOK (predicts alone but redundant once the dominant prior is controlled), or REFUTED. This operationalises Venet et al. [10]: a signature is a novel mechanism only if it survives adjustment for the dominant prior. On Luminal A vs B against the proliferation anchor (Figure 2A), the basal/keratinization hypothesis is SUPPORTED ($\Delta +0.039$, $\beta = 8$), an immune/cytotoxic hypothesis is REFUTED (AUROC 0.51, adds 0.000), and a random 30-gene set is EXPLAINED_BY_TEXTBOOK (predicts at 0.63 alone but adds ≈ 0). Scaled to a whole library (rank_hypotheses over the 50 MSigDB Hallmark sets; Figure 2B), the screen is self-validating: the proliferation-type hallmarks (E2F, G2M, MYC) add exactly 0 beyond the anchor (EXPLAINED — they *are* the anchor’s axis), while the SUPPORTED hits are the orthogonal lineage programs led by estrogen-response (early & late) — matching the known LumA/B biology of proliferation plus ER signalling.

A binary verdict can mislead when a hypothesis fails to add: *absent* and *redundant-because-*

collinear look identical (both $\Delta \approx 0$) yet mean opposite things. We therefore attach a gate-free commonality decomposition (after Tonidandel & LeBreton [11]) to every hypothesis: the variance it explains is partitioned into the part **unique** beyond the anchor versus the part **common** (shared with it), and its effect is split into direct versus anchor-**mediated**, yielding a `collinearity_label` $\in \{\text{NOVEL}, \text{REDUNDANT}, \text{INERT}\}$. This makes the one cross-cohort non-reproduction in our screen honest rather than misleading (Figure 2C): the estrogen-response hypothesis is SUPPORTED in TCGA but not in METABRIC — yet the decomposition shows the difference is *structural, not biological*. In TCGA, ER carries variance unique beyond proliferation (unique $R^2 = 0.038 \rightarrow \text{NOVEL}$); in METABRIC the same ER signal is essentially entirely shared with proliferation (redundancy = 1.00, 96 % of its effect mediated through the proliferation anchor $\rightarrow \text{REDUNDANT}$), because LumB’s lower ER is collinear with its higher proliferation in that cohort. ER biology is *redundant* in METABRIC, not *absent* — a distinction the raw verdict cannot make and a transportability caveat for any anchored screen ported across cohorts. A controlled sweep makes the mechanism explicit (Figure 2D): holding both marginal effects fixed and varying *only* the anchor-hypothesis correlation, the same ER effect is labelled NOVEL in 100 % of simulated cohorts at TCGA’s correlation (+0.19) but falls into a collinear/suppression valley at METABRIC’s (−0.17, $\approx 2\%$ NOVEL). The verdict is thus governed by each cohort’s covariate distribution — a generalisability/transportability property [12] — which is why an anchored hypothesis screen should be re-characterised, not merely re-run, across cohorts. Repeating the analysis on a second endpoint (HER2, with the ERBB2-amplicon anchor and an ER hypothesis) confirms the labels are specific rather than a single “fails to add”: there the ER $\rightarrow$ HER2 marginal effect itself differs across cohorts, so HER2 is *not* a same-marginal flip — ER is **INERT** in TCGA (Cohen’s $d \approx -0.05$, too weak to be novel or redundant) but **REDUNDANT** in METABRIC ($d \approx -0.29$, redundancy 0.89, 66 % mediated by the amplicon). Across the two endpoints the framework cleanly separates all three regimes — *novel* (ER in TCGA Luminal A/B), *redundant/collinear* (ER in METABRIC Luminal A/B and HER2), and *absent/weak* (ER for HER2 in TCGA).

Extending this to a four-endpoint $\times$ two-cohort panel (Figure 3) makes the practical payload explicit: of four endpoints, two transport and two do not. The transportable pairs are orthogonal, biologically robust axes — a basal/keratinization anchor with an immune hypothesis is NOVEL in *both* cohorts (immune infiltration adds a real axis beyond the basal lineage program), and an ER-signature anchor with a proliferation hypothesis carries a small but genuine unique proliferation slice in both. The two that do *not* transport are exactly the ER-collinearity cases above (proliferation $\rightarrow$ ER on Luminal A/B; amplicon $\rightarrow$ ER on HER2). The transportability caveat is therefore specific and predictable: it attaches to hypotheses whose collinearity with the anchor is itself cohort-dependent, while genuinely anchor-orthogonal axes transport cleanly.

3. Discussion

The method instantiates and unifies three established traditions — clinical-offset / incremental-value modelling (an established model as offset, omics on the residual; the closest methodological twin) [5]; biologically-informed models that bake known biology into structure [6,7]; and machine learning on established gene signatures [8]. Its specific contribution is a packaged, never-below-anchor, margin-gated residual on a *zero-parameter* prior, plus a residual-discovery framing with an explicit noise control, and a demonstration that the discovered axes’ external reproducibility is predictable from their biological coherence. The honest scope: predictive gains over a strong anchor are modest — the value is *routing and discovery*, not large AUROC wins — consistent with the rarity of genuine super-additive multi-omics gains [1,2]. Discovered axes are candidate hypotheses; the basal axis is externally validated, the HER2 axis is cohort-specific, and other endpoints await further cohorts.

Anchored hypothesis labels across 4 endpoints × 2 cohorts

	TCGA	METABRIC
LumA_vs_LumB proliferation→ER ✗differs	NOVEL EXPLAINED_B red=-4.95	REDUNDANT REFUTED red=1.00
HER2_pos_vs_neg ERBB2_amplicon→ER ✗differs	INERT EXPLAINED_B red=-7.80	REDUNDANT EXPLAINED_B red=0.89
ER_pos_vs_neg ER_signature→proliferation ✓transports	NOVEL EXPLAINED_B red=0.89	NOVEL EXPLAINED_B red=0.96
Basal_vs_rest basal→immune ✓transports	NOVEL SUPPORTED red=0.45	NOVEL SUPPORTED red=0.48

■ NOVEL ■ REDUNDANT ■ INERT

Figure 1: **Figure 3. Anchored hypothesis labels across four endpoints × two cohorts.** Each cell is one (anchor → hypothesis) test in one cohort, coloured by the commonality label (NOVEL = carries variance unique to the hypothesis beyond the anchor; REDUNDANT = predicts but collinear/mediated; INERT = no appreciable signal), with the cross-validated verdict and the redundancy below it. Rows are endpoints (anchor → hypothesis); ✓transports/✗differs marks whether the label is concordant across TCGA and METABRIC. Basal→immune and ER-status→proliferation transport (NOVEL in both); the two ER-collinearity endpoints (Luminal A/B, HER2) do not.

4. Methods

Anchored integration. `select_anchor` scores each modality by repeated stratified-CV AUROC (tie-broken by robustness) and picks the top composite, inside the outer CV. `anchored_integrate` pins the anchor and adds a secondary on its residual as $\text{logit}(\text{anchor}) + \beta \cdot \text{secondary}$, with $\beta \geq 0$ chosen on held-out data ($\beta = 0$ allowed; a margin gate plus repeated inner CV control small-sample noise). `forward_integrate` / `auto_integrate` extend this to any number of modalities by greedy forward selection. **Knowledge anchor.** `signature_score` builds a fixed mean-z score from a curated gene set; `knowledge_anchored_integrate` pins it as the anchor. **Residual discovery.** `anchored_residual_discovery` ranks features by partial correlation with the label controlling for the anchor, retains those orthogonal to the anchor ($|r| < 0.6$), gates the panel, and tests it against matched random panels and (for verification) held-out splits and permuted labels. Because the matched-random-panel null saturates when an endpoint is broadly predictable (e.g. squamous-vs-adeno, ER status), the method also reports a **selection-stability** statistic — the recurrence of the discovered panel across 50% subsamples relative to a permuted-label null — which stays informative there (lung basal axis stability gain $\approx +0.89$; genome-wide ER methylation $\approx +0.63$, both where the panel-vs-random null is uninformative). Data: TCGA-BRCA (Xena HiSeqV2, HumanMethylation450, clinical) and METABRIC (microarray, clinical). Pathway enrichment: g:Profiler. All functions are in `omniomics.multiomics`; runners and recorded metrics are in the repository. **Scalability.** The discovery is linear and streams: a vectorized residualize-then-correlate, an optional one-pass Sure-Independence-Screening pre-filter, Benjamini-Hochberg FDR, and parallel nulls let it run on genome-wide feature matrices out of core. As an end-to-end check, screening all 485,577 HumanMethylation450 probes ($n = 829$) to the top 5,000 by association with ER status took ≈ 24 s in a single low-memory pass; the ESR1-methylation anchor (AUROC 0.65) then rose to 0.90 with genome-wide CpGs, and the top CpG beyond ESR1 mapped to PGR (the canonical ER-coreregulated gene) — textbook biology recovered at scale.

5. Data and Code Availability

Code, runners, recorded metrics, unit tests and continuous-integration guards: the `omniomics` repository (`reports/dmoi_*.py`, `*_results.csv`, `tests/`). Key runners: `dmoi_knowledge_anchor.py`, `dmoi_residual_discovery.py`, `dmoi_discovery_{er,her2}.py`, `dmoi_external_{metabric,her2_metabric}.py`, `dmoi_external_lung.py`, `dmoi_external_hnsc.py`, `dmoi_discovery_nsclc_io.py`, `dmoi_clinical_survival.py`, `dmoi_meth_genomewide.py` (genome-wide scale). Scale utilities: `omniomics.scale`, `omniomics.representations`. Full methods note: `reports/anchored_integration_methods.md`. All data are public and de-identified: TCGA (BRCA, LUAD, LUSC, HNSC) via UCSC Xena; METABRIC via cBioPortal; the NSCLC anti-PD-1 cohort from Hellmann et al. (MSK 2018) via cBioPortal.

Declarations

Competing interests. The author declares no competing interests.

Funding. No external funding was received for this work.

Author contributions. H.R.K. conceived the method, performed all analyses, and wrote the manuscript.

Ethics. This study uses only publicly available, de-identified data from established repositories (TCGA/UCSC Xena, cBioPortal); no new human-subjects data were collected, so no additional ethics approval was required.

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AI assistance. Analyses and drafting were carried out with the assistance of an AI coding agent under the author's direction; all results are reproducible from the cited runners.

Figure

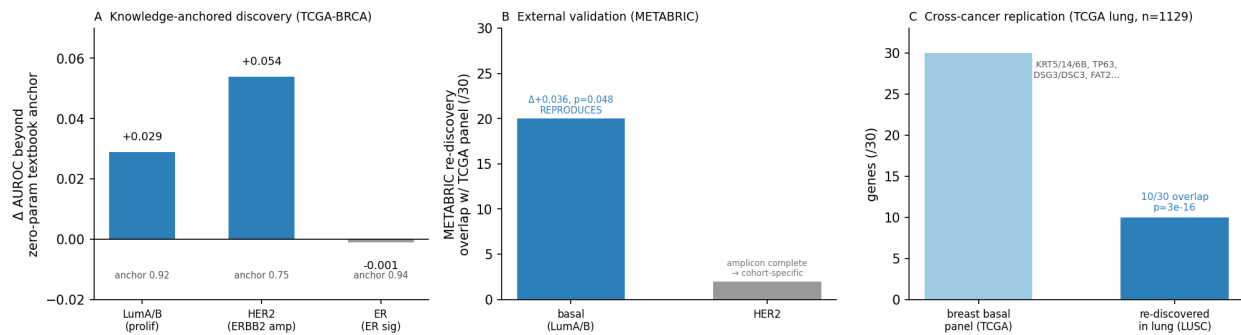


Figure 2: Figure 1. Knowledge-anchored discovery and its external validation. (A) Across three TCGA-BRCA endpoints, the gain in AUROC from adding genome-wide data beyond a zero-parameter textbook anchor: the data adds a real axis where the textbook is incomplete (LumA/B proliferation → basal, +0.029; HER2 amplicon → neuroendocrine/immune, +0.054) and nothing where it is complete (ER signature, -0.001; specificity). **(B)** In the independent METABRIC cohort, an unbiased re-discovery recovers the basal axis (20/30 overlap with the TCGA panel; direct replication $\Delta +0.036$, $p = 0.048$) but not the HER2 axis (2/30), because the amplicon anchor is already complete in METABRIC. **(C)** Cross-cancer replication: in TCGA lung (LUAD vs LUSC, $n = 1,129$) the same proliferation-anchored residual re-discovers the squamous/keratinization program, overlapping the breast basal panel 10/30 (KRT5/14/6B, TP63, DSG3/DSC3, FAT2...; hypergeometric $p \approx 3 \times 10^{-16}$) — the discovered biology, not a cohort artefact.

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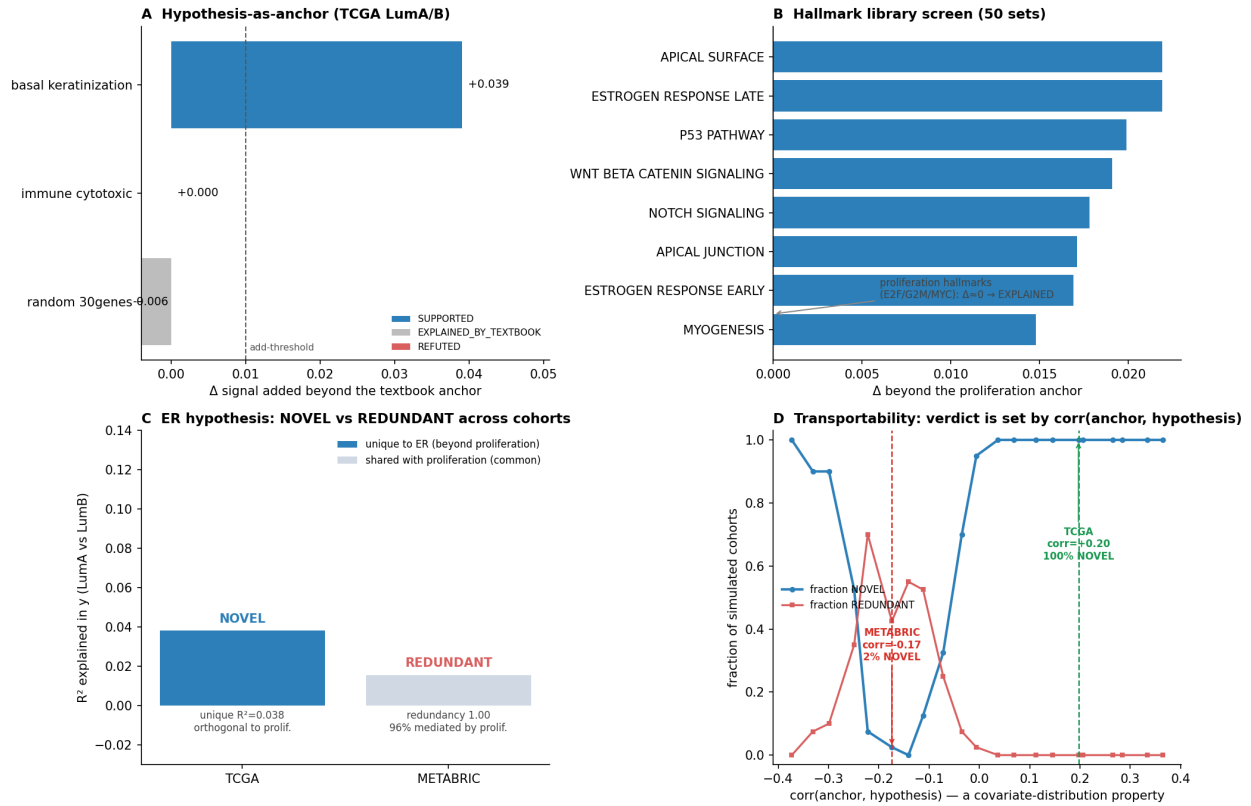


Figure 3: Figure 2. Hypothesis-as-anchor: confirm, explain-away, or refute. (A) On TCGA-BRCA Luminal A vs B, three hypotheses tested against the textbook proliferation anchor by the signal each adds beyond it: the basal/keratinization hypothesis is SUPPORTED ($\Delta +0.039$), an immune/cytotoxic hypothesis is REFUTED (adds 0.000), and a random 30-gene set is EXPLAINED_BY_TEXTBOOK (predicts alone but adds ≈ 0 once proliferation is controlled). (B) Screening all 50 MSigDB Hallmark gene sets as candidate hypotheses: the proliferation-type hallmarks (E2F, G2M, MYC) add exactly 0 beyond the anchor (EXPLAINED — the same axis), while the SUPPORTED hits are the orthogonal lineage programs led by estrogen-response, matching known Luma/B biology. (C) Commonality/mediation re-characterization of the estrogen-response hypothesis across cohorts: stacked R^2 in the Luma/B endpoint split into the part *unique* to ER (beyond proliferation, blue) versus the part *shared* with proliferation (grey). ER is NOVEL in TCGA (unique R^2 0.038, orthogonal to proliferation) but REDUNDANT in METABRIC (redundancy 1.00, 96 % mediated through proliferation) — collinear, not absent. (D) Transportability of the verdict: holding both marginal effects fixed and varying *only* the anchor-hypothesis correlation, the fraction of simulated cohorts labelled NOVEL vs REDUNDANT is plotted against that correlation; the two real cohorts sit on the curve at their measured corr(proliferation, ER) — the *same* ER effect is 100 % NOVEL at TCGA's +0.19 but collapses into the collinear/suppression valley at METABRIC's -0.17. The verdict is a covariate-distribution property, not a difference in ER biology.

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